Analysis and Classification of Tremors in a Parkinson's Disease Simulator Using Machine Learning

Erick Toque
Biomechanics and Applied Robotics Research Laboratory
Pontificia Universidad Católica del Perú
Lima, Peru
estoque@pucp.edu.pe

Sebastian Vila
Biomechanics and Applied Robotics Research Laboratory
Pontificia Universidad Católica del Perú
Lima, Peru
vila.s@pucp.edu.pe

César Gutiérrez-Flores
Biomechanics and Applied Robotics Research Laboratory
Pontificia Universidad Católica del Perú
Lima, Peru
cesar.gutierrezf@pucp.edu.pe

Rosa M. Silva-Salas
Biomechanics and Applied Robotics Research Laboratory
Pontificia Universidad Católica del Perú
Lima, Peru
rosam.silva@pucp.edu.pe

Victoria E. Abarca
Biomechanics and Applied Robotics Research Laboratory
Pontificia Universidad Católica del Perú
Lima, Peru
victoria.abarca@pucp.edu.pe

Dante A. Elias
Biomechanics and Applied Robotics Research Laboratory
Pontificia Universidad Católica del Perú
Lima, Peru
delias@pucp.pe

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***Abstract*—Hand tremor is a symptom typically observed in patients with Parkinson's disease (PD). However, repetitive hand movements in healthy patients or non-PD conditions can also be confused with this symptom. In that sense, this work develops a systematic analysis for differentiating the types of tremors in reference using Machine Learning techniques and a tremor simulator mechanism equipped with inertial sensors that will provide the necessary dataset for such analysis. According to scientific literature, this mechanism is based on frequency analysis with a principal component of about 5 Hz. The results show that the best classification model is the random forest with favorable metrics such as an accuracy of 98.66% and an F1 score of 98.66%. This will allow a classification of the nature of tremors for subsequent application in diagnosing PD, reducing complexity in the clinical analysis through data collection with inertial sensors and applying an optimized algorithm. In addition, it means a step forward in automating clinical procedures to benefit patients with this type of disease with symptomatological particularities, such as hand tremors.**

***Index Terms*—classification, hand, machine learning, tremor, Parkinson.**

I. Introduction

Parkinsonian tremor (PT) is often confused in symptoms with essential tremor (ET), two different forms of motor disorders, each with some manifestations of its concerning physical activity. While PT tends to occur more frequently during rest periods, ET occurs more commonly during the execution of actions [1]. The overlap of symptoms between both disorders hinders their accurate diagnosis, leading to approximately 25% of Parkinson's disease patients being misdiagnosed as having essential tremors. This results in an inadequate allocation of medical resources [2]. The hanging hand position is the most suitable for evaluating the amplitude of resting tremor in Parkinson's disease (PD) [3]. There are several studies where significant differences between ET and PT can be analyzed. By using accelerometers placed on the back of each hand and employing multiscale entropy, the physiological complexity of hand tremor can be analyzed [1]. On the other hand, tremors in patients with Parkinson's disease tend to be more localized and less complex than in individuals without neurological disorders [4]. Wrist movements of pronation-supination and abduction-adduction are more common in PT and less repetitive in healthy individuals, as subclinical tremor in Parkinson's disease has significantly higher amplitude and frequency in certain specific movements than typical tremor [5]. Regarding classification, recurrent neural networks based on BiLSTM can accurately differentiate between PT, ET, and individuals without disorders, using accelerometer data from the lower arm, hand, and upper arm, achieving an 80% accuracy [2]. Additionally, there are cases where patients with a long history of ET who later develop PD exhibit a synchronous pattern of resting tremor, unlike those who only have PD. These patients also retain more significant tremors during posture or movement and lower axial and limb rigidity scores compared to patients with PD predominated by tremor [6]. Additionally, it has been observed that patients who first have ET and later develop PD could be a particular group, with causes that interact in the early stages of the disease [7]. Moreover, patients with ET have a higher risk of developing PD over their lifetime compared to those without ET [8]. Resting tremors in patients with ET can be a late manifestation, an independent disorder, indicate progression to PD, or be a misdiagnosis [9]. In addition, some patients

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mimic Parkinson's tremors, which further complicates the differential diagnosis. Although they present with PD-like symptoms, they do not have the disease. Given this complexity, our study uses machine learning algorithms to differentiate between PD-like tremors, voluntary tremors, and no tremors. We aim to improve diagnostic accuracy and enable early interventions for patients with Parkinsonian symptoms. Our method offers an improved quantification of hand tremors, explicitly addressing a gap in the Latin American context, where such studies are scarce. Given the low frequency of consultations for patients with Parkinson's disease, typically every 3-6 months [15], it is crucial to have an effective monitoring system outside the clinical setting. This method will allow for continuous monitoring, facilitating neurologists to make more informed treatment decisions and potentially improving patient outcomes. Machine learning techniques can be employed to achieve the stated goals since they facilitate pattern recognition, dataset description, and data classification. For example, Meng et al. [10] use a Support Vector Machine (SVM) to classify healthy and Parkinson's patients based on hand tremors with an accuracy of 91%. Similarly, Sun et al. [11] use the frequency domain and an SVM to achieve a 92% F1 score in a cross-validation. Likewise, Tong et al. [12] has a study of hand tremors using inertial sensors where, thanks to a convolutional neural network (CNN), it achieves an accuracy of 97.32%. Other authors, such as LeMoyne et al. [13], used a wearable inertial sensor and managed to use a multilayer perceptron neural network with an accuracy of 95%. Overall, in an exploratory search of IEEE Xplore, 17 papers presented at conferences and three journal publications between 2022 and 2024 were identified using our keywords as a filter.

II. Material and Method

The Knowledge Discovery in Databases (KDD) methodology with a supervised learning paradigm is used to classify whether the acquired signals are Parkinson's tremor data or healthy patients. First, the acquisition of signals, then a pre-processing of the data, and we assigned an appropriate label. After that, the features in the time and frequency domain are extracted. Once the data is understood, the most relevant features are selected, tests are made with classification models, and finally, based on comparable metrics, the final model is chosen. Also, it's important to mention that we used a simulator to generate the database for this study. Each sample took 10 seconds to obtain, for a total of 903 samples. The selected sample rate, 70 Hz, complies with the sampling theorem by being higher than twice the 42 Hz Nyquist frequency since, in this case, 21 Hz was considered an acceptable value for the maximum limit of the frequency analysis, given that the frequency of interest is approximately 5 Hz. This process is also represented in the Fig. 1.

A. Signal Acquisition

The tremor simulator data collection process was performed using an MPU6050 inertial sensor with three accelerometers and three gyroscopes (one for each axis), as shown in Fig. 2.

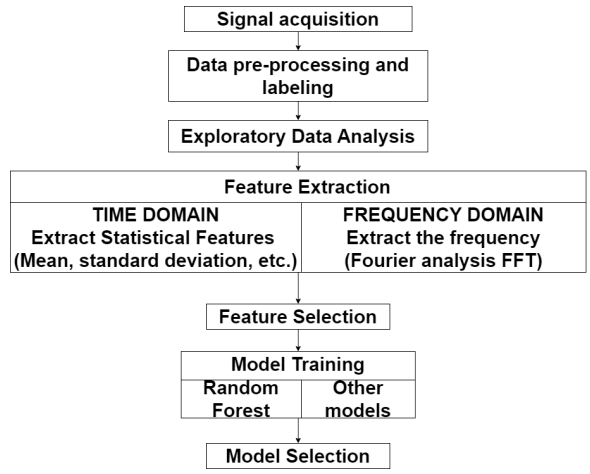

Fig. 1. Process Flow Diagram of The Knowledge Discovery in Databases.

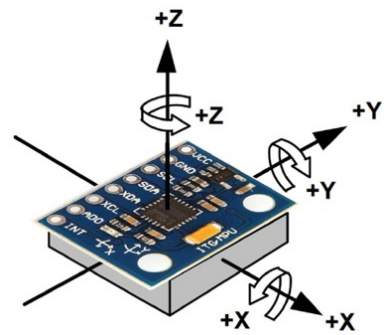

Fig. 2. Triaxial acceleration and angular velocity axes of MPU6050.

The simulator mechanism was configured to obtain three movements: a tremor movement with a fundamental frequency of 5 Hz related to Parkinson's disease, a voluntary tremor movement with a fundamental frequency of 3 Hz, and the absence of a hand tremor. The movements were acquired from a simulator; for simplicity of use, we will refer to these as Parkinson's tremor, voluntary tremor, and no tremor. It obtains the characteristics of 6 signs of the sensor, which are the following:

- ID: Sample number (simulated patient identifier)
- T: Time in milliseconds
- A_X: Acceleration of the x-axis of MPU6050 in m/s^2
- A_Y: Acceleration of the y-axis of MPU6050 in m/s^2
- A_Z: Acceleration of the z-axis of MPU6050 in m/s^2
- G_X: Rotation of the x-axis of MPU6050 in °/s
- G_Y: Rotation of the y-axis of MPU6050 in °/s
- G_Z: Rotation of the z-axis of MPU6050 in °/s

The time variable synchronized the data obtained, and the identifiers helped extract the characteristics. To better understand the data obtained, the sensor must be located at point G and have the orientation of its axes as indicated in detail A. In addition, we have control over rotation at point C with a stepper motor, the distance at which the supports are placed, the CD radio, and the distance at which the G point

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is placed to the support points, as shown in Fig. 3. This is achieved thanks to an oscillatory movement generated by the slider crank mechanism.

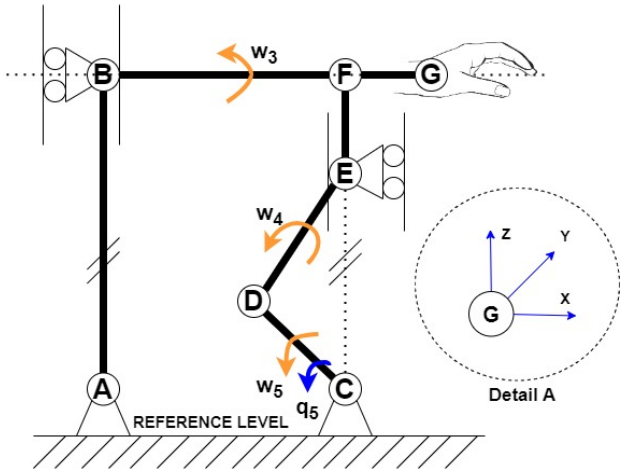

Fig. 3. Mechanism of the tremor simulator and location of the inertial sensor in the simulator.

B. Data Preprocessing and labeling

Once the samples are obtained, the data is cleaned and adjusted to avoid working with empty, duplicate, or atypical values. For the preprocessing and labeling, we found the following, considering that we work with CSV files:

- Invalid values such as numerical symbols or alphanumeric strings are removed.
- Last row with incomplete information is removed: accelerations and angular velocities are obtained from the sensor; if at least one is missing, the row is removed.
- A supervised Machine Learning approach is used where simulated Parkinson's patient data is labeled as ParkinsonTremor, simulated healthy patient data is labeled as NoTremor, and simulated non-Parkinson's tremor data is labeled as VoluntaryTremor.

After that process, the Fast Fourier Transform (FFT) and a 2 to 21 Hz band pass filter were also applied to these signals. Filtering helps us cancel the offset value, which is irrelevant as it is the magnitude of a frequency of 0 Hz, making spectral analysis difficult. Limiting the frequency range to a range suitable for frequency analysis is essential in this work.

C. Exploratory data analysis

The data are displayed in the time and frequency domain for a random sample of patients intended for training. In the time domain, we have the acceleration and angular velocity graphs between the first and third seconds to see if there is a pattern. It was applied to Parkinson's tremor, voluntary tremor, and absence of tremor. But more emphasis was given to how the hand-tremor-related graphs could be differentiated whether or not they were voluntary, as shown in Fig. 4 and 5.

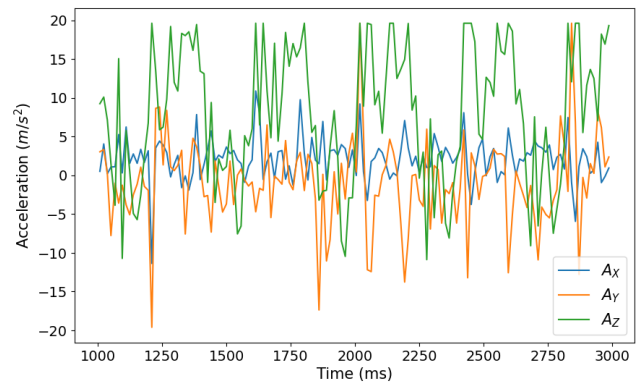

Fig. 4. Accelerations in the time domain of the simulation of a patient without Parkinson's.

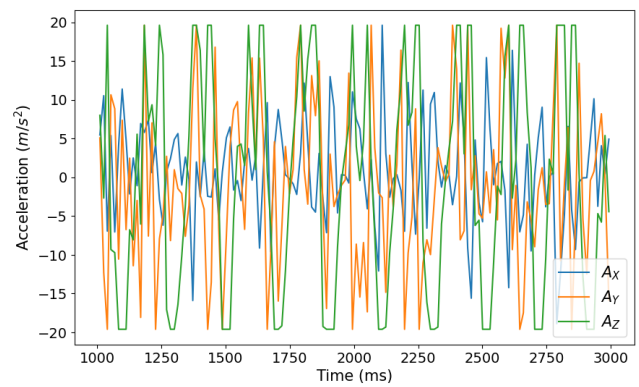

Fig. 5. Accelerations in the time domain of the simulation of a patient with Parkinson's.

The accelerations of a patient with Parkinson's disease have more frequency than expected. Still, it needs to be distinguishable, and this becomes more difficult when you have a more significant time window. However, this sometimes appears true in the time domain as depicted in Fig. 6 and 7. Furthermore, the rotation of the X-axis does not seem important in the physical implementation, but its graphics tend to be cyclic.

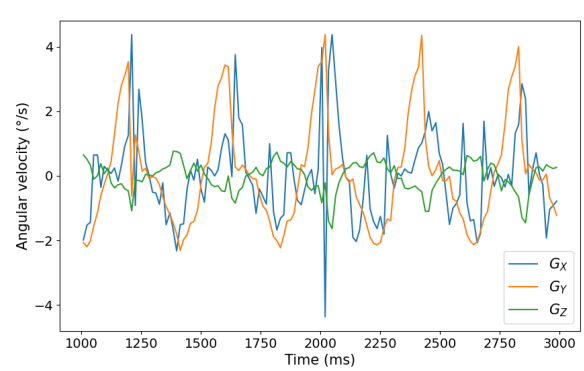

Fig. 6. Angular velocity in the time domain of the simulation of a patient without Parkinson's.

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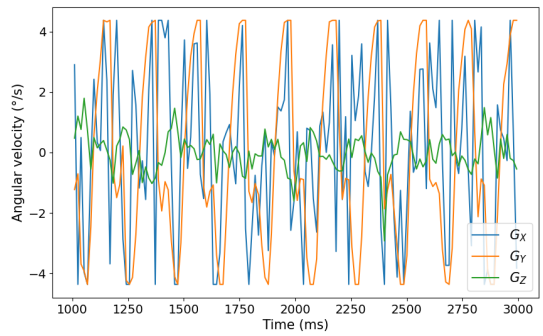

Fig. 7. Angular velocity in the time domain of the simulation of a patient with Parkinson's.

In addition, considering the literature reviewed, plotting the frequency domain where the Parkinsonian movement has a fundamental frequency of 5 Hz for the accelerations was decided. The first two peaks and the frequencies at which these peaks occur were extracted from the frequency domain. All of the above was applied for the three accelerations and rotational speeds. Fig. 8 shows that the fundamental frequency is 3 Hz, and the magnitude corresponding to this frequency is greater than 3. In the case of a patient without tremor, the magnitude is of the order of thousandths. In Fig. 9, it is depicted that we can observe notable small magnitudes accompanied by high acceleration in the Y and Z axes.

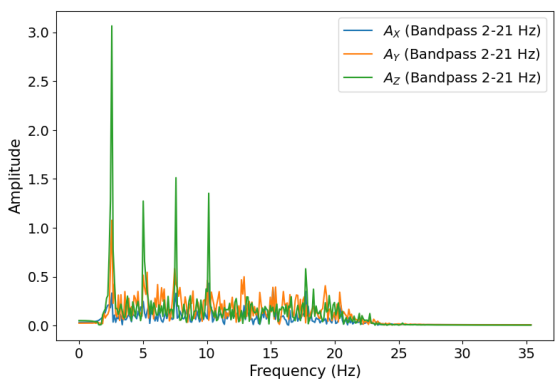

Fig. 8. Frequency-domain voluntary tremor accelerations in a healthy patient.

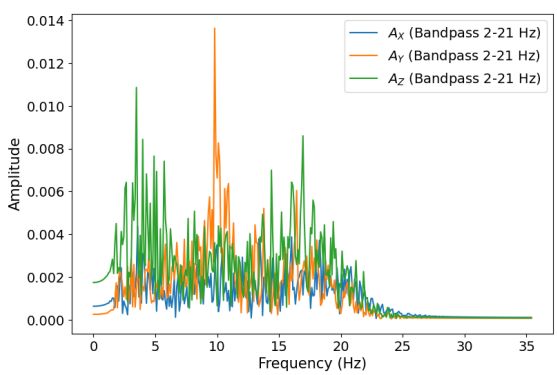

Fig. 9. Accelerations of a healthy patient without hand tremor in the frequency domain.

A Parkinson's tremor is depicted in Fig. 10, exhibiting the expected fundamental frequency of 5 Hz and a magnitude greater than 6.

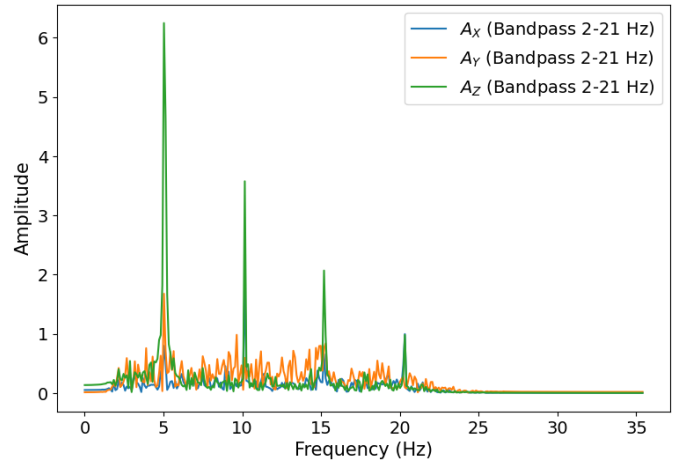

Fig. 10. Frequency-dominant hand tremor accelerations in a Parkinson's disease patient.

D. Feature Extraction

First, the data had to be analyzed in the time domain; in this case, the average of the descriptors of all the samples obtained was used. Therefore, the mean, median, first and third quartiles, standard deviation, maximum, and minimum were extracted. Below are three tables belonging to the databases of healthy patients and patients with Parkinson's disease. These tables present the accelerations and angular velocity in the three axes, highlighting this database's standard deviation and first and third quartile. When analyzing the data presented in the three tables, we observed distinctive patterns in the turning and acceleration measurements among the different patient groups.

Table I represents patients without tremors and shows that the turning values are significantly low, suggesting a more stable motor activity. Furthermore, the equality between the mean and median values indicates a symmetrical distribution of the data, while the low standard deviation indicates minimal variability in these measurements.

TABLE I
DESCRIPTORS OF CENTRAL TENDENCY AND DISPERSION FOR THE PATIENTS WITHOUT TREMORS

Statistical Measures	**Accelerations and angular velocity**					
	A_X	A_Y	A_Z	G_X	G_Y	G_Z
Mean	-0.72	1.20	9.82	-0.04	0.01	-0.03
Standard deviation	0.07	0.12	0.08	0.01	0.00	0.01
Median	-0.72	1.20	9.82	-0.04	0.01	-0.03
First quartile	-0.75	1.13	9.78	-0.04	0.01	-0.03
Third quartile	-0.68	1.26	9.86	-0.03	0.01	-0.02

In contrast, in Table II, which shows patients with tremors, we observed higher values of twist and accelerations, especially in the A_X direction. The similarity between the

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mean and median values suggests a similar data distribution. However, the high standard deviation in accelerations indicates significant variability, which could reflect the erratic nature of tremor movements. In Table III, representing patients with Parkinson's disease, the turning values remain low, but the acceleration measurements show significant differences compared to the previous tables. The discrepancy between the mean and median values suggests a skewed distribution of the data, and the high standard deviation indicates considerable variability, which could be characteristic of the involuntary movements associated with the disease.

TABLE II
CENTRAL TENDENCY AND DISPERSION DESCRIPTORS FOR THE PATIENTS WITH VOLUNTARY TREMOR

Statistical Measures	**Accelerations and angular velocity**					
	A_X	A_Y	A_Z	G_X	G_Y	G_Z
Mean	2.13	0.14	7.95	-0.09	0.01	-0.01
Standard deviation	3.29	6.43	8.56	1.60	1.59	0.59
Median	2.10	0.22	9.13	-0.16	-0.08	0.07
First quartile	0.79	-3.09	1.43	-1.22	-1.15	-0.48
Third quartile	3.39	3.55	14.89	0.79	0.61	0.48

TABLE III
DESCRIPTORS OF CENTRAL TENDENCY AND DISPERSION FOR THE PATIENTS WITH PARKINSON'S TREMOR.

Statistical Measures	**Accelerations and angular velocity**					
	A_X	A_Y	A_Z	G_X	G_Y	G_Z
Mean	0.24	0.74	2.38	-0.05	0.03	-0.04
Standard deviation	6.47	10.85	14.71	2.67	3.10	0.64
Median	0.47	-0.34	3.72	0.25	-0.68	-0.11
First quartile	-2.61	-5.48	-11.21	-2.18	-2.52	-0.51
Third quartile	3.57	7.66	19.56	1.77	3.37	0.44

In Tables IV and V, we have the first two harmonics represented by frequencies 1 and 2 (f_1 and f_2) of each signal captured by the sensor. Of the 903 patients, the mean was taken by grouping by label, and by each indicator, it is observed that the frequency of A_Z is 5 Hz and that the frequency of G_Y is also 5 Hz.

TABLE IV
CAPTURED FREQUENCIES OF ACCELERATIONS FOR SIMULATED PATIENTS.

Statistical Measures	**Frequencies (mean)**					
	$A_X f_1$	$A_X f_2$	$A_Y f_1$	$A_Y f_2$	$A_Z f_1$	$A_Z f_2$
Parkinson Tremor	10.22	20.91	5.13	15.98	5.05	15.97
Voluntary Tremor	2.87	15.45	2.57	14.73	2.57	17.40
NoTremor	6.21	17.23	9.85	20.40	6.74	17.61

TABLE V
CAPTURED FREQUENCIES OF ANGULAR VELOCITY FOR SIMULATED PATIENTS.

Statistical Measures	**Frequencies (mean)**					
	$G_X f_1$	$G_X f_2$	$G_Y f_1$	$G_Y f_2$	$G_Z f_1$	$G_Z f_2$
Parkinson Tremor	5.26	16.13	5.05	16.06	5.06	15.82
Voluntary Tremor	2.57	15.25	2.57	17.26	2.57	15.85
No Tremor	9.76	20.32	7.53	18.18	9.64	20.18

E. Model Training

We used 67% of the data for training and 33% for testing the trained model containing the three classes. A total of 301 simulations were run for each class: voluntary tremor, no hand tremor, and Parkinson's tremor.

Random Forest is a machine learning algorithm combining multiple decision trees' output to achieve a single result. The random forest algorithm is an extension of the bagging method, as it uses both bagging and feature randomness to create an uncorrelated decision tree forest. Some benefits of random forest are reduced risk of overfitting, flexibility, and ability to determine characteristic importance [14]. Also, there are other classification models, such as the k-nearest neighbor algorithms (KNN), Gradient boosting, multinomial logistic regression, and artificial neural networks (ANN). Other models were in slope, but they were presented to those who had better metrics.

F. Model selection

The model selection is based on four main metrics: accuracy, precision, recall, and F1-score. The metrics are calculated as weighted averages over the three classes, that is, in the functions used by Scikit-Learn. Metrics indicate that the frequency of each class is considered when calculating the average. Good metrics are sought to rank the model well and minimize false positives and negatives.

III. RESULTS

In this section, we present the outcomes of our study, which encompass the performance metrics of various machine learning models in classifying Parkinsonian and non-Parkinsonian tremors. Additionally, we provide a detailed analysis using confusion matrices to evaluate the efficacy of the best model in distinguishing between different tremor types.

A. Model comparison

The model with better accuracy, precision, recall, and F1-score is slightly better for the Random forest model, although the difference is slight, as depicted in Table VI. Generally, the models with lower performance are the multinomial logistic regression and the ANN. The most representative metrics chosen are accuracy and F1-score.

TABLE VI
COMPARISON OF THE METRICS OF THE CLASSIFICATION MODELS.

Classification Models	**Metrics**			
	Accuracy	***Precision***	***Recall***	***F1-score***
Random Forest	0.9866	0.9871	0.9866	0.9866
KNN	0.9865	0.9870	0.9865	0.9865
Gradient boosting	0.9865	0.9870	0.9865	0.9865
Logistic regression	0.8255	0.8619	0.8255	0.8046
ANN	0.8825	0.9005	0.8825	0.8748

In addition, the model chosen for having better metrics had the fundamental frequency of acceleration in Z and angular speed in Y as central features, as detailed in Table VII.

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TABLE VII
TOP SELECTED FEATURES AND THEIR IMPORTANCE

Feature	Importance
AZ_peak1_freq	0.4347
GY_peak1_freq	0.3674
GY_peak2_freq	0.1561
AZ_peak2_freq	0.0418

B. Confusion Matrix

Fig. 11 represents the confusion matrix, which shows a high performance of the classification model Random Forest. In the main diagonal, we observe high values, indicative of accurate classification in most classes. Errors are mainly concentrated in classifying tremors without Parkinson's and voluntary movements. However, these errors do not represent a significant problem, as they do not substantially affect the overall model performance.

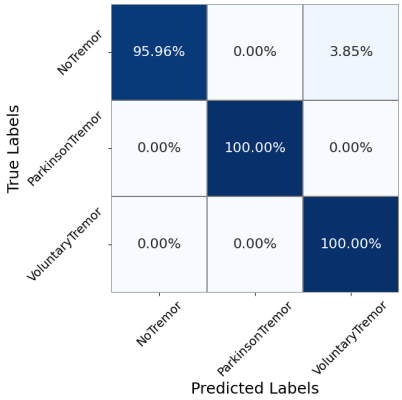

Fig. 11. Random forest model normalized confusion matrix.

IV. CONCLUSION

The study showed that the Random Forest model outperformed others in the tremor classification, achieving an accuracy of 98.66% and an F1 score of 98,66%. This was achieved by considering both the statistical and graphical analysis in the time domain and the frequency of the signals collected by the inertial sensor. It is suggested that the simulator be configured to simulate different hand sizes and better fit the characterization of voluntary tremors and Parkinson's tremors to create a more robust model with better classification. In addition, we note that including velocity and displacement measurements derived from accelerations could further improve the model's performance. These findings suggest promising avenues for future research in diagnosing and managing diseases such as Parkinson's disease by analyzing hand tremors and earlier and more accurate detection.

V. FUTURE WORKS

Future work includes collecting signals from patients with Parkinson's tremors, essential tremors, and healthy patients. In addition, information such as age, gender, and general health status will be considered. This would also help better capture tremor variations in actual patients. On the other hand, it is intended to improve the simulator, expand the tremor analysis considering additional parameters such as frequency ranges and the calibration of inertial sensors, and develop medical technology for diagnosis and monitoring based on this work.

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